

# US Patent & Trademark Office

## Patent Public Search | Text View

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United States Patent Application Publication

20250285747

Kind Code

A1

Publication Date

September 11, 2025

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### PATIENT SAFETY DEVICE MAINTENANCE SYSTEM

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#### Abstract

A method may include receiving, from a safety system coupled to an infusion system, repair data associated with a repair event, during which a user accesses the infusion system. The method may also include detecting, based on the repair data, the repair event is unauthorized. The method may also include causing, based on the detecting, at least one safety adjustment to operation of the infusion system. Related methods and articles of manufacture, including apparatuses and computer program products, are also disclosed.

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**Family ID:** 1000008626479

**Appl. No.:** 18/724523

**Filed (or PCT Filed):** December 29, 2021

**PCT No.:** PCT/US2021/065546

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#### Publication Classification

**Int. Cl.:** G16H40/40 (20180101); A61M5/142 (20060101)

**U.S. Cl.:**

**CPC** G16H40/40 (20180101); A61M5/142 (20130101);

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#### Background/Summary

##### TECHNICAL FIELD

[0001] The subject matter described herein relates generally to a device maintenance system and

more specifically to a patient safety device maintenance system for a medical device such as an infusion system.

## BACKGROUND

[0002] Infusion devices, such as infusion pump systems, can include a complicated arrangement of components. Repairing and/or maintaining the infusion devices can be difficult. Also, an improperly repaired and/or assembled infusion device may render the device unusable or cause injury to a patient. Thus, maintenance personnel or other users who perform maintenance or repairs on the infusion devices should be qualified, trained, and/or certified to ensure the infusion devices are assembled or repaired properly. However, it can be difficult to ensure that only authorized personnel make repairs and/or perform maintenance on the infusion devices, and to track both authorized and unauthorized repairs and/or maintenance. Furthermore, it can be difficult to prevent and/or track unauthorized repairs or use of unauthorized replacement parts, which can similarly render the infusion devices unusable, damage components of the infusion devices, or lead to injury to the patient.

## SUMMARY

[0003] Systems, methods, and articles of manufacture, including computer program products, are provided for a patient safety device maintenance system for an infusion system. The patient safety device maintenance system consistent with implementations of the current subject matter may assist with tracking maintenance of a medical device (e.g., an infusion system) and adjust the medical device in response to detection of an unauthorized access of the medical device.

[0004] According to some aspects, a method includes receiving, from a safety system coupled to an infusion system, repair data associated with a repair event, during which an element within a housing of the infusion system is accessed. The method may also include detecting, based on the repair data, the repair event is unauthorized. The method may also include causing, based on the detecting, at least one safety adjustment to operation of the infusion system.

[0005] In some aspects, the repair event includes one or more of a repair of the element of the infusion system, a replacement of the element of the infusion system, opening the housing, and a maintenance performed on the infusion system.

[0006] In some aspects, the infusion system includes an infusion pump.

[0007] In some aspects, the infusion system includes: an infusion pump, and a control device coupled to the infusion pump. The control device includes at least one data processor and at least one memory storing instructions which, when executed by the at least one data processor, control operation of the infusion pump.

[0008] In some aspects, the safety system includes a tamper switch. The tamper switch includes: at least two electrically conductive surfaces of the infusion system. Contact between the at least two electrically conductive surfaces forms a circuit. The tamper switch may also include a fastener configured to maintain contact between the at least two electrically conductive surfaces. Removal of the fastener is configured to allow access to the infusion system during the repair event. Removal of the fastener is further configured to activate the tamper switch.

[0009] In some aspects, the safety system further includes the housing of the infusion system. The housing defines at least one of the at least two electrically conductive surfaces.

[0010] In some aspects, the repair data includes information indicating activation of the tamper switch. The detecting includes determining the repair event is unauthorized when the tamper switch is activated.

[0011] In some aspects, the safety system includes a damage switch configured to indicate the infusion system should be accessed during the repair event.

[0012] In some aspects, the damage switch includes one or more of a moisture sensor configured to be activated upon detecting an ingress of fluid into an interior of the infusion system and an accelerometer configured to be activated upon detecting an acceleration indicating an impact on the infusion system.

[0013] In some aspects, the repair data includes information indicating activation of one or more of the moisture sensor and the accelerometer.

[0014] In some aspects, the safety system includes: a supervisor chip coupled to the infusion system, a primary switch coupled to the supervisor chip and to the infusion system, and a secondary switch coupled to the supervisor chip and to the infusion system.

[0015] In some aspects, the primary switch includes the tamper switch, and the secondary switch includes the damage switch.

[0016] In some aspects, the repair data includes information indicating activation of one or more of the primary switch and the secondary switch.

[0017] In some aspects, the at least one safety adjustment includes recording, by the supervisor chip, a timestamp associated with the activation of the one or more of the primary switch and the secondary switch.

[0018] In some aspects, the at least one safety adjustment includes erasing a security value from memory of the supervisor chip, thereby limiting use of the infusion system.

[0019] In some aspects, the method includes resetting the security value upon receipt of a selection from an authorized user, thereby restoring the use of the infusion system.

[0020] In some aspects, the safety system includes a tamper evident data tag. The repair data includes no detection of the tamper evident data tag, and the detecting includes determining the repair event is unauthorized based on the no detection of the tamper evident data tag.

[0021] In some aspects, the safety system includes a light sensor and/or switch within a housing of the infusion system. The repair data includes a scan of a memory coupled to the light sensor and/or switch to determine whether an event indicating the light sensor and/or switch was triggered is stored in the memory. The light sensor and/or switch may be triggered when the housing is removed from the infusion system, and the detecting includes determining the repair event is unauthorized based on a determination that the event is stored in the memory.

[0022] In some aspects, the safety system includes a camera coupled to the housing of the infusion system. The repair data includes a light level in a field of view of the camera. The detecting includes determining the repair event is unauthorized based on the light level in the field of view of the camera corresponding to a light level threshold.

[0023] In some aspects, the safety system includes a drop sensor coupled to the infusion system. The repair data includes a pattern of movement reflecting a repair of the infusion system, and the detecting includes determining the repair event is unauthorized based on the pattern of movement.

[0024] In some aspects, the system include an authorized user registry, and the repair data includes user data associated with the infusion system during the repair event. Determining the repair event is unauthorized includes determining that the user data is unassociated with a user included in the authorized user registry.

[0025] In some aspects, the system an authorized user registry, and the repair data includes user data associated with the infusion system during the repair event. Determining the repair event is unauthorized includes determining that the user data is associated with a user authorized to perform a different repair event than the repair event in the authorized user registry.

[0026] In some aspects, the unauthorized user is one or more of unqualified, not trained, and not certified.

[0027] In some aspects, the safety system includes a location sensor coupled to the infusion system. The repair event is unauthorized when the repair event is performed in an unauthorized location detected by the location sensor.

[0028] In some aspects, the at least one safety adjustment includes a transmission of one or more of a visual alert, an audio alert, an audiovisual alert, and a tactile alert.

[0029] In some aspects, the audio alert includes one or more of a periodic noise and an audio file indicating the infusion system was repaired by an unauthorized user.

[0030] In some aspects, the at least one safety adjustment includes a display of an indication via a

user interface coupled to the infusion pump. The indication indicates the infusion system was repaired by an unauthorized user.

[0031] In some aspects, the at least one safety adjustment includes a display of an indication via a user interface coupled to the infusion pump. The indication includes at least one of a banner and a persistent and perceivable indicator.

[0032] In some aspects, the at least one safety adjustment includes disabling network connectivity in the infusion system.

[0033] In some aspects, the at least one safety adjustment includes transmission, to a device management server coupled to the infusion system, of a message indicating the repair event was unauthorized.

[0034] In some aspects, the at least one safety adjustment includes creation, in a distributed ledger, of an entry includes an identifier associated with the infusion system and a timestamp of the repair event.

[0035] In some aspects, the distributed ledger is a blockchain ledger.

[0036] In some aspects, the entry further includes an identifier associated with the user.

[0037] In some aspects, the at least one safety adjustment includes storage of a security value in a secure register of the infusion system. The security value indicates one or more of the detecting and a timestamp of the repair event.

[0038] In some aspects, the user includes one or more of a manufacturer, a repair personnel, a clinician, and a patient.

[0039] According to some aspects, an infusion system includes: a supervisor chip, a primary switch coupled to the supervisor chip, a secondary switch coupled to the supervisor chip, and a controller including at least one data processor and a memory for storing instructions, which when executed by the at least one data processor, result in operations. The operations include recording, by the supervisor chip, a timestamp associated with activation of the one or more of the primary switch and the secondary switch. The activation indicates unauthorized access of the infusion system.

[0040] In some aspects, the operations include erasing a security value from memory of the supervisor chip, thereby preventing use of the infusion system.

[0041] In some aspects, the operations further include: resetting the security value upon receipt of a selection from an authorized user, thereby allowing the use of the infusion system.

[0042] In some aspects, the primary switch is a tamper switch includes: at least two electrically conductive surfaces. Contact between the at least two electrically conductive surfaces forms a circuit. The tamper switch may also include a fastener configured to maintain contact between the at least two electrically conductive surfaces. Removal of the fastener is configured to allow access to the infusion system. Removal of the fastener is further configured to activate the tamper switch.

[0043] In some aspects, the safety system further includes a housing of the infusion system. The housing defines at least one of the at least two electrically conductive surfaces.

[0044] In some aspects, the operations include determining the access to the infusion system is unauthorized when the tamper switch is activated.

[0045] In some aspects, the secondary switch is a damage switch configured to indicate the infusion system should be accessed.

[0046] In some aspects, the damage switch includes one or more of a moisture sensor configured to be activated upon detecting an unwanted ingress of fluid into an interior of the infusion system and an accelerometer configured to be activated upon detecting an acceleration indicating an impact on the infusion system.

[0047] Implementations of the current subject matter can include methods consistent with the descriptions provided herein as well as articles that comprise a tangibly embodied machine-readable medium operable to cause one or more machines (e.g., computers, etc.) to result in operations implementing one or more of the described features. Similarly, computer systems are also described that may include one or more processors and one or more memories coupled to the

one or more processors. A memory, which can include a non-transitory computer-readable or machine-readable storage medium, may include, encode, store, or the like one or more programs that cause one or more processors to perform one or more of the operations described herein. Computer implemented methods consistent with one or more implementations of the current subject matter can be implemented by one or more data processors residing in a single computing system or multiple computing systems. Such multiple computing systems can be connected and can exchange data and/or commands or other instructions or the like via one or more connections, including, for example, to a connection over a network (e.g. the Internet, a wireless wide area network, a local area network, a wide area network, a wired network, or the like), via a direct connection between one or more of the multiple computing systems, etc.

[0048] The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims. While certain features of the currently disclosed subject matter are described for illustrative purposes in relation to a patient safety device maintenance system, it should be readily understood that such features are not intended to be limiting. The claims that follow this disclosure are intended to define the scope of the protected subject matter.

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## Description

### DESCRIPTION OF DRAWINGS

[0049] The accompanying drawings, which are incorporated in and constitute a part of this specification, show certain aspects of the subject matter disclosed herein and, together with the description, help explain some of the principles associated with the disclosed implementations. In the drawings,

[0050] FIG. 1 depicts a system diagram illustrating a patient safety device maintenance system, in accordance with some example implementations;

[0051] FIG. 2 schematically depicts an example infusion system, in accordance with some example implementations;

[0052] FIG. 3A depicts an example infusion system, in accordance with some example implementations;

[0053] FIG. 3B depicts an example infusion system, in accordance with some example implementations;

[0054] FIG. 4A depicts a flowchart illustrating a process for safe repair of an infusion device, in accordance with some example implementations;

[0055] FIG. 4B depicts a flowchart illustrating a process for improving patient safety, in accordance with some example implementations;

[0056] FIG. 5 depicts a block diagram illustrating a computing system, in accordance with some example implementations;

[0057] FIG. 6A depicts a front view of a patient care system, in accordance with some example implementations;

[0058] FIG. 6B depicts an enlarged view of a portion of a patient care system, in accordance with some example implementations;

[0059] FIG. 6C depicts a perspective view of a pump, in accordance with some example implementations;

[0060] FIG. 7 depicts a process diagram for secure repair of a device, in accordance with some example implementations; and

[0061] FIG. 8 depicts a state diagram of user interfaces for improving patient safety when using a repaired infusion device, in accordance with some example implementations.

[0062] When practical, similar reference numbers denote similar structures, features, or elements.

## DETAILED DESCRIPTION

[0063] Infusion devices, such as infusion pump systems, can include a complicated arrangement of components. Repairing and/or maintaining the infusion devices can be difficult. Thus, maintenance personnel or other users who perform maintenance or repairs on the infusion devices should be qualified, trained, and/or certified to ensure the infusion devices are assembled or repaired properly. However, in many instances, patients or other users (e.g., medical facilities, medical personnel, and/or the like) may choose to repair the infusion devices on their own or by using an unauthorized third party. Using an unauthorized third party or improperly repairing the device may render the device unusable or lead to an injury to a patient using the device.

[0064] As an example, some infusion devices require specialized equipment and fixtures to repair the devices and/or perform maintenance on the devices. For example, an infusion device may include a series of springs to control occluders, and/or the like. To repair such an infusion device, a fixture may be used to hold the series of springs in place when a housing of the infusion device is removed, and one or more components of the infusion device are repaired and/or replaced. If the fixture is not used, the springs or other components may become loose or become bent, resulting in jamming of the springs and/or complete failure of the infusion device. Both instances can lead to injury to the patient due to the infusion device malfunctioning as a result of the improperly assembled and/or damaged infusion device.

[0065] The patient safety device maintenance system consistent with implementations of the current subject matter helps to prevent, limit, or track such issues. For example, the patient safety device maintenance system consistent with implementations of the current subject matter helps to detect when an infusion device has been modified by an unauthorized user and, upon detection of the unauthorized use, make one or more safety adjustments at the infusion device to protect patient and clinical safety. Thus, the patient safety device maintenance system may protect the integrity of the infusion device and help to ensure safe operation of the infusion device.

[0066] In some instances, manufacturers of the infusion devices may authorize certain authorized users or third parties to make repairs and/or perform maintenance on the infusion devices. As noted above, the maintenance personnel who perform maintenance or repairs on the infusion devices may be qualified (e.g., trained, and/or certified) to ensure the infusion devices are assembled or repaired properly in a manner that meets or exceeds the required quality standards. However, it may be difficult to ensure that the maintenance personnel performing the maintenance or repairs is actually qualified. The patient safety device maintenance system consistent with implementations of the current subject matter helps to track the personnel (authorized and/or unauthorized) performing the maintenance and/or repairs on the infusion devices. Additionally or alternatively, the patient safety device maintenance system consistent with implementations of the current subject matter helps to ensure that only qualified personnel (e.g., authorized users) are able to perform maintenance and/or repairs on the infusion devices.

[0067] Furthermore, unauthorized replacement parts can cause critical failure of the infusion devices and/or cause the infusion devices to operate in a way that is harmful to the patient using the infusion devices. As an example, unauthorized keypads can significantly impact operation of the infusion devices and control over the delivery of medication to the patient using the infusion devices. As another example, unauthorized batteries or other materials may not be subjected to rigid quality control procedures before being implemented in the infusion devices, possibly leading to damage to the infusion devices and/or harm to the patient.

[0068] Despite the potential harms associated with using an unauthorized replacement part in an infusion device and/or an unauthorized repair of the infusion device, it can be difficult to prevent and/or track unauthorized repairs or use of unauthorized replacement parts. To determine whether the infusion devices have been opened or repaired, some infusion devices employ paint or other markers to seal the device. However, such paint or markers can be easily faked and/or broken.

Infusion devices also generally do not have mechanisms for tracking the use of replacement parts and/or maintenance performed on the infusion devices. Thus, it can be difficult to ensure that only authorized personnel make repairs and that those personnel only use authorized replacement parts. The patient safety device maintenance system consistent with implementations of the current subject matter helps to determine when an infusion device has been accessed, and to track the access of the infusion device. The patient safety device maintenance system may also perform one or more safety adjustments to the operation of the infusion device to improve the safety of the patient and to help prevent future unauthorized access of the infusion device.

[0069] FIG. 1 depicts a system diagram illustrating a patient safety device maintenance system **100**, consistent with implementations of the current subject matter. Referring to FIG. 1, the patient safety device maintenance system **100** may include a safety engine **110**, an infusion system **120**, a display **154**, a client device **99**, and a data system **125**. The safety engine **110**, the display **154**, and/or the data system **125** may form a portion of the infusion system **120** and/or may be positioned within a housing (e.g., a housing **270**) of the infusion system **120**.

[0070] The infusion system **120** may include a pump **122** (also referred to herein as an “infusion device”). Additionally or alternatively, the infusion system **120** may include a pump control device coupled to and/or forming a part of the pump **122**. The pump control device may include at least one processor and at least one memory for storing instructions, which when executed by the at least one processor cause the pump **122** to perform one or more operations, such as to deliver a medication to a patient. The pump **122** may be a TCI pump, a syringe pump, an anesthesia delivery pump, a patient-controlled analgesic (PCA) pump, a large volume pump (LVP), a small volume pump (SVP), an ambulatory pump, and/or the like, and may be configured to deliver a medication to a patient. However, it should be appreciated that the pump **122** may be any infusion device configured to deliver a substance (e.g., fluid, nutrients, medication, and/or the like) to a patient's circulatory system or epidural space via, for example, intravenous infusion, subcutaneous infusion, arterial infusion, epidural infusion, and/or the like. Alternatively, the pump **122** may be an infusion device configured to deliver a substance (e.g., fluid, nutrients, medication, and/or the like) to a patient's digestive system via a nasogastric tube (NG), a percutaneous endoscopic gastrostomy tube (PEG), nasojejunal tube (NJ), and/or the like. Moreover, the pump **122** may be part of a patient care system (e.g., a patient care system **20**) that includes one or more additional pumps.

[0071] The display **154** may form a part of the infusion system **120** (e.g., the pump **122**) or may be separately coupled as part of a client device **99**. The display **154** may also include a user interface. The user interface may form a part of a display screen of the display **154** that presents information to a user and/or the user interface may be separate from the display screen. For example, the user interface may be one or more buttons, or portions of the display screen that is configured to receive an entry from the user. As described herein, the display **154** may additionally or alternatively display one or more alerts or other information associated with a repair event, during which a user may access the infusion system **120** (e.g., the pump **122**) to repair, replace, maintain, check, update, or clean one or more elements (e.g., device elements such as parts, components, or the like) of the system **120**. The access during the repair event may be virtual such as to update or maintain firmware of the device or the access may be physical such as to open the housing of the infusion system **120** to perform work on a battery or other element in the housing. The repair event is generally performed while the infusion system **120** is unassociated with a patient or specific infusion order.

[0072] The client device **99** may be a mobile device such as, for example, a smartphone, a tablet computer, a wearable apparatus, and/or the like. However, it should be appreciated that the client device **99** may be any processor-based device including, for example, a desktop computer, a laptop computer, a workstation, and/or the like. For example, via the client device **99**, the user may be able to configure certain parameters of the pump **122**, such as an air in line threshold, a rate limit, an alarm limit, and the like. Additionally, in some examples, via the client device **99**, the user may

configure various drug protocols with default settings and safety parameters (e.g., setting a limit to a dose of a drug).

[0073] The data system **125** may include one or more databases, providing physical data storage within a dedicated facility and/or being locally stored on the pump **122** and/or the display **154**. The data system **125** may include an inventory system, a patient care system, an administrative system, an electronic medical record system, and/or the like, which store a plurality of electronic medical records, each of which include the patient's medical history, one or more fluid delivery protocols, and/or the like. Additionally and/or alternatively, the data system **125** may include cloud-based systems providing remote storage of data in, for example, a multi-tenant computing environment and/or the like. The data system **125** may also include non-transitory computer readable media.

[0074] In some implementations, the data system **125** may include and/or be coupled to a server **126** (e.g., a device management server), which may be a server coupled to a network, a cloud server, and/or the like. The client device **99** may wirelessly communicate with the server **126**. The server **126** may communicate directly with the pump **122**, with the pump **122** via the safety engine **110**, and/or through the client device **99** and/or the display **154**. The server **126**, which may include a cloud-based server, may provide data and/or instructions from the data system **125** to the pump **122** and/or the safety engine **110**, the client **99**, or the display **154** to be transmitted to the pump **122**. Additionally or alternatively, the server **126** may receive data, such as repair data (e.g., data associated with a repair event, during which a user accesses the infusion system **120**, such as the pump **122**) from the pump **122** and/or the safety engine **110**. The data system **125** may include an authorized user registry. The authorized user registry may include user data that identifies users and permissions for performing repair events. The registry may include a binary list of users that are authorized. The registry may include a binary list of users that are unauthorized. The registry may associate specific users with specific repair events (e.g., battery replacement, door replacement, software upgrade), devices (e.g., pump model, brand, version, type, etc.), or device elements (e.g., battery, housing, pump module, etc.). The registry may be indexed by user identifier to expedite the identification of appropriate registry entries for a specific user.

[0075] The safety engine **110**, the pump **122**, the display **154**, the client device **99**, and/or the data system **125** may be communicatively coupled via a network **105**. Meanwhile, the network **105** may be any wired and/or wireless network including, for example, a public land mobile network (PLMN), a local area network (LAN), a virtual local area network (VLAN), a wide area network (WAN), the Internet, and/or the like.

[0076] The safety engine **110**, which may define a controller, may include at least one processor and/or at least one memory for storing instructions, which when executed by the at least one processor performs one or more operations to, for example, improve a safety of a patient and/or track access of the infusion system **120** (e.g., the pump **122**). The safety engine **110** may form a part of and/or be coupled to the pump **122**.

[0077] In some implementations, the safety engine **110** receives repair data from a safety system **204** (see FIG. 2) of the infusion system **120** (e.g., the pump **122**). The repair data may include data associated with a repair event, during which a user accesses the infusion system **120**. The user includes one or more of a patient, a clinician, a device manufacturer, a repair personnel, and/or the like. The user accesses the infusion system **120**, such as the pump **122** during a repair event that includes one or more of a repair of an element (e.g., device elements such as parts, components, or the like) of the infusion system **120**, maintenance performed on the infusion system **120**, cleaning of the infusion system **120**, performing a mechanical check on the infusion system **120**, and/or the like. For example, during the repair event, the user may open the pump **122**, remove the housing **270** of the pump **122**, and/or remove one or more other components of the pump **122**.

[0078] The user accessing the infusion system **120** should be authorized as an authorized user. In other words, the user accessing the infusion system **120** should be trained and/or certified for making repairs and/or performing maintenance on the infusion system **120**. If the user is not



authorized (e.g., is an unauthorized user), the user may not properly repair the infusion system, perform maintenance on the infusion system **120**, use the infusion system **120**, and/or the like, which may lead to failure of the infusion system **120** and/or harm to the patient using the infusion system **120**.

[0079] To improve the safety of the patient, the patient safety device maintenance system **100** (e.g., via the safety engine **110**) may detect when an unauthorized user accesses the infusion system and/or when a user uses an unauthorized component during the access of the infusion system. For example, the safety engine **110** may detect the repair event is unauthorized based on the repair data received from the safety system **204**. The repair event is unauthorized when the unauthorized user accesses the infusion system **120** and/or uses an unauthorized component in the infusion system **120**.

[0080] In some implementations, when the safety engine **110** determines that the repair event is unauthorized, the safety engine **110** may cause at least one safety adjustment to operation of the infusion system **120**. The safety adjustment may include transmission of an alert, storage of information associated with the repair event, transmission of an alarm, disablement of the infusion system **120**, changing an operation of the infusion system **120**, and/or the like. Thus, the safety engine **110** may track access of the infusion system **120** for later auditing and/or investigation, and in some instances, may prevent use of the infusion system **120** until the infusion system **120** is evaluated by an authorized user.

[0081] For example, the safety adjustment may include transmission of one or more of a visual alert, an audio alert, an audio visual alert, a tactile alert, and/or another alert. The audio alert may include a periodic noise and/or an audio file indicating the infusion system was repaired and/or accessed by an unauthorized user. The safety adjustment may additionally or alternatively include a display of an indication, via a user interface (e.g., of the display **154** and/or a display **236**), indicating the infusion system was accessed by an unauthorized user. The safety adjustment may additionally or alternatively include a display of an indication including at least one banner or persistent and perceivable indicator. The safety adjustment may additionally or alternatively include disabling network connectivity in the infusion system **120**. The safety adjustment may additionally or alternatively include transmission (e.g., to the safety engine **110** or directly to the server **126**), of a message indicating the repair event was authorized and/or unauthorized. The safety adjustment may additionally or alternatively include creation, in a distributed ledger (e.g., a blockchain ledger) of the infusion system **120**, of an entry including an identifier associated with the infusion system, a timestamp of the repair event, and/or an identifier associated with the user accessing the infusion system **120**. The safety adjustment may additionally or alternatively include storage of a security value in a secure register of the infusion system indicating the detection of the unauthorized repair event and/or a timestamp associated with the repair event.

[0082] FIG. 2 schematically depicts an example of the infusion system **120** (e.g., the pump **122**), consistent with implementations of the current subject matter. As shown in FIG. 2, the infusion system **120** may include the safety engine **110**, a housing **270**, a user interface **130**, a data storage **272** (which may be the same as and/or include the data system **125**), a safety system **204**, and communications circuitry **226**. While the infusion system **120** is described as including the safety system **204**, the safety engine **110**, the user interface **130**, the housing **270**, the communications circuitry **226**, and the storage **272**, it should be appreciated that the safety system **204**, the safety engine **110**, the user interface **130**, the housing **270**, the communications circuitry **226**, and the storage **272** may be implemented in one or more other medical devices or medical systems, such as a medication preparation station, medication dispensing stations, flow cytometers, and other medical instruments.

[0083] Referring to FIG. 2, the user interface **130** may include a badge reader **238**, a biometrics scanner **240**, a display **236**, a user input **234**, and/or the like. The user interface **130** may be coupled to or integrated with the pump **122**. In some implementations, at least a portion of the user interface

**130** forms a part of the pump **122**, and at least a portion of the user interface **130** is coupled to an external client device, such as a computer, mobile phone, and/or the like, which is communicatively coupled to the pump **122**. In some implementations, the user interface **130** forms a part of, includes, and/or is coupled to the display **154**.

[0084] The user interface **130** may receive data that is used for a later audit of the repair events at the pump **122**. For example, the pump **122** may include one or more auditing features. The one or more auditing features may be features that allow for the repair events to be tracked and associated with a user, such as the user accessing the pump **122**. For example, the pump **122** may record information collected when the pump **122** is accessed, including the identification tag (barcode, RFID tag, etc.) of the user and/or the identity of the user who accessed the pump **122**, videos recorded during the repair event, and physical property measurements taken during the repair event. The user interface **130** may provide the badge reader **238** for reading an identification code of the user and/or the biometric scanner **240** for obtaining biometric features of the user. The identification code of the user and/or the biometric features of the user may be received by the user interface **130** and be stored as a part of a record. The record may be linked to or associated with the user for tracking and later auditing. The record may also include time and date details to associate timing with the repair event.

[0085] In some implementations, the user interface **130** includes a display **236**. The display **236** may display an alert, such as a visual alert, an audio alert, an audiovisual alert, a tactile alert, and/or the like. The alert may indicate information about the repair event, such as when the repair event is unauthorized. The audio alert, for example, may include a periodic noise or an audio file indicating the infusion system was accessed by an unauthorized user. In some implementations, the display **236** displays an indication including a banner and/or a persistent and perceivable indicator relaying information about the repair event, such as when the repair event was unauthorized.

[0086] The user interface **130** may also include a user input **234**. The user input **234** may include dials, buttons, icons, selectable areas, or other perceivable indicia presented via the user interface **130** that, when interacted with (e.g., clicked, touched, selected, etc.), initiates an exchange of data for the infusion system **120** to present via the user interface **130**. For example, the user input **234** may receive information about the repair event, about the user, and/or a secure value for securing the infusion system **120**.

[0087] Referring again to FIG. 2, the infusion system **120** may include wired or wireless communications circuitry **226** that is connected to and/or controlled by the safety engine **110**. The communication circuitry **236** may include one or more antennas (which may include a near-field communication (NFC) antenna, a Bluetooth antenna, a Bluetooth Low Energy antenna, a Wi-Fi antenna, and/or other antennas), a barcode reader, and/or a radio frequency identification (RFID) tag reader. The antenna may be configured to read from and/or write to a data tag positioned on or otherwise coupled to the infusion system **120**, such as the pump **122**. The data tag may be a type of wireless transceiver and may include a microcontroller unit (MCU), a memory, and an antenna (e.g., an NFC antenna) to perform the various functionalities described herein. The data tag may be, for example, a 1 Kbit or a 2 Kbit NFC tag. NFC tags with other specifications may also be used.

[0088] Additionally, and/or alternatively, the infusion system **120** may include communications circuitry **236** including one or more cellular communications features, such as via the network **105**. The one or more cellular communications features may include General Packet Radio Services (GPRS), Long-Term Evolution (LTE) networks, 5G cellular technology, and/or the like. Such cellular communications features may increase system mobility and deployment options.

[0089] The communications circuitry **236** may include additional components/circuitry for other communication modes, such as, for example, NFC circuitry, Bluetooth circuitry, Bluetooth Low Energy circuitry, and/or Wi-Fi circuitry and associated circuitry (e.g., control circuitry), for communication with other devices. For example, the pump **122** may be configured to wirelessly communicate with a remote processor (e.g., the display **154**, a smartphone, a tablet, wearable

electronics, a cloud server, and/or the like) through the communications circuitry **236**, and through this communication may receive and/or transmit one or more fluid parameters, one or more patient parameters, one or more clinician parameters, one or more fluid delivery protocols, and/or the like from and/or to one or more of the remote processors. In some implementations, based on the detection of an unauthorized repair event, the safety engine **110** may disable the communications circuitry **236**. In other implementations, based on the detection of an unauthorized repair event, the safety engine **110** may cause the communications circuitry **236** to transmit information (e.g., the repair data, a timestamp, an alert, etc.) representing the repair event to a medical facility, the data system **125**, and/or the like.

[0090] In some embodiments, the infusion system **120** includes a data storage **131**, including one or more databases, data tables, and/or the like. The data storage **131** may store (e.g., temporarily store) the repair data, a timestamp of the repair event, a user identifier of the user accessing the infusion system, a device identifier associated with the infusion system, and/or the like. The data storage **131** may include a distributed ledger, such as a blockchain ledger.

[0091] In some implementations, one or more components of the infusion system **120**, such as the replacement components of the infusion system **120** being added to the infusion system, may be scanned, such as by the infusion system **120** (e.g., the pump **122**). For example, if a door or another component of the pump **122** is replaced, the repair personnel may, during the repair event, take one or more repair actions, including activating a repair mode on the pump **122**, scanning the new door (or other component), making the repair, and/or verifying the repair is complete. One or more identifiers associated with the repair actions can be logged and/or written to the distributed ledger. This allows for the stored identifiers and information relating to the repair event to be evaluated later, such as during an audit. This configuration also helps to track unauthorized repair events at the pump **122**. In some implementations, after and/or before the identifiers are logged to the distributed ledger, the safety engine **110** may request that the user scan and/or present authorization credentials, such as via the badge reader **238**, biometrics scanner **240**, and/or the like. The scanned credentials may be additionally or alternatively be stored in the distributed ledger.

[0092] Referring again to FIG. 2, the infusion system **120** includes a safety system or device **204**. The safety system **204** may include at least one of a tamper switch **206**, a damage switch **208**, a supervisor chip **214**, a tamper evident data tag **216**, a light sensor **218**, a camera **220**, a drop sensor **222**, a location sensor **224**, and/or the like. The safety system **204** may be coupled to and/or form at least a part of the pump **122**.

[0093] FIGS. 3A and 3B illustrate an example of the infusion system **120** including the tamper switch **206**, consistent with implementations of the current subject matter. As shown in FIGS. 3A and 3B, the pump **122** includes the housing **270**. The housing **270** may surround one or more internal components of the pump **122**, including one or more internal structures. The housing **270** may be secured to the one or more internal structures via a fastener **302**, such as a mechanical fastener (e.g., a screw), and/or the like. Loosening and/or removal of the fastener **302** may allow for removal of the housing **270** and, thus, access to the one or more internal structures of the pump **122**. The fastener **302** may include one, two, three, four, five, ten or more fasteners.

[0094] The fastener **302** may maintain contact between at least two electrically conductive surfaces of the pump **122**. For example, the at least two electrically conductive surfaces may include a first electrically conductive surface **304** and a second electrically conductive surface **306**. In some implementations, the first electrically conductive surface **304** includes at least a portion of the housing **270** and the second electrically conductive surface **306** includes at least a portion of one of the one or more internal structures of the pump **122**.

[0095] Contact between the at least two electrically conductive surfaces **304**, **306** may form a circuit. For example, when the at least two electrically conductive surfaces **304**, **306** are in contact with one another, the circuit may be closed. Alternatively, when the at least two electrically conductive surfaces **304**, **306** are not in contact with one another, the circuit may be opened.

Opening of the circuit may activate the tamper switch **206**, which may include and/or be coupled to the at least two electrically conductive surfaces **304**, **306**, the fastener **302**, and/or the housing **270**. [0096] Thus, when the fastener **302** is removed and/or loosened, such as during a repair event, the one or more internal structures of the pump **122** may be accessed, and the tamper switch **206** may be activated. In this configuration, the repair data received by the safety engine **110** may include information indicating the activation of the tamper switch. Based on the information indicating the activation of the tamper switch, the safety engine **110** may determine the repair event is unauthorized, and as a result, may cause at least one safety adjustment described herein.

[0097] Referring back to FIG. 2, the safety system **204** may include the damage switch **208**. The damage switch **208** may indicate the infusion system should be accessed during a repair event, such as when the damage switch **208** detects an impact and/or damage caused to the pump **122**. For example, the damage switch **208** may include a moisture sensor **210**, an accelerometer **212**, and/or the like.

[0098] The moisture sensor **210** may be activated upon detecting an unwanted ingress of fluid into an interior of the pump **122** and/or infusion system **120**. The moisture sensor **210** may thus determine that the pump **122** has experienced fluid ingress and may need to be evaluated or repaired before being used by a patient. The moisture sensor **210** may include a film based sensor with multiple conducting lines. In some implementations, the moisture sensor **210** can include or be coupled to a circuit that includes an absorbent pad and the multiple conducting lines. When moisture enters the pump **122**, the moisture may be captured on an absorbent pad. The absorbent pad capturing the moisture can close the circuit, indicating that a fault has occurred, and activating the moisture sensor (or damage switch **208**). Using the moisture sensor, the repair data received by the safety engine, and from the moisture sensor, can include information indicating the activation of the moisture sensor.

[0099] The accelerometer **212** may detect an acceleration of the pump **122**. For example, the accelerometer **212** can determine whether the pump **122** has been dropped and/or experiences an acceleration that exceeds a threshold acceleration, indicating a drop of the pump **122**. A drop of the pump **122** may cause damage to the pump **122**. Thus, the accelerometer **212** readings may indicate whether the pump **122** experienced damage or potentially experienced damage, and as a result should be evaluated. The acceleration exceeding the threshold acceleration may activate the accelerometer **212** (or damage switch **208**). Using the accelerometer, the repair data received by the safety engine, and from the accelerometer, can include information indicating the activation of the accelerometer.

[0100] In some implementations, the damage switch **208** (or the tamper evident data tag **216**, the light sensor **218**, the camera **220**, the drop sensor **222**, the location sensor **224** or other safety systems) may be used in conjunction with the tamper switch **206**, to detect the repair event is unauthorized. For example, the safety system **204** may include the supervisor chip **214**. The supervisor chip **214** may include a processor (e.g., a microprocessor) and a memory (e.g., random access memory and/or the like). The supervisor chip **214** may help to ensure proper operation of the pump **122** and/or to detect unauthorized repair events.

[0101] For example, the supervisor chip **214** may include at least one, two, or more inputs that are connected to one or more other safety systems **204** (e.g., the tamper switch **206**, the tamper evident data tag **216**, the light sensor **218**, the camera **220**, the drop sensor **222**, the location sensor **224** or other safety systems). When the supervisor chip **214** determines that one or both of the safety systems **204** coupled to the supervisor chip **214** are activated, the supervisor chip **214** may record a timestamp indicating when the safety systems **204** were activated. The timestamp may be recorded in the memory of the supervisor chip **214** and/or in the data storage **272**. The timestamp can be used to later audit and/or investigate a repair event that occurred at the time the safety systems **204** were activated.

[0102] In some implementations, the supervisor chip **214** may store a security value (e.g., a

numeric value, an alphanumeric value, and/or the like) in the memory of the supervisor chip **214**. The security value may include a serial number of the pump **122** and/or may be otherwise associated with the pump **122**. In this configuration, in order for the pump **122** to be used, the safety engine **110** may query the supervisor chip **214** for the security value. The safety engine **110** may allow use of the pump **122** when the safety engine **110** receives the security value in response to the query. When at least one, two, or more safety systems **204** are activated, the supervisor chip **214** may erase the security value from the memory of the supervisor chip **214**.

[0103] In order for the pump **122** to be placed back into service, the pump **122** would need to be evaluated and/or repaired by an authorized user, who can reset the security value stored in the memory of the supervisor chip **214**. For example, the safety engine **110** may detect receipt of a selection by a user via the user input **234** of an option to reset the security value and/or receipt of a new security value. The safety engine **110** may, based on the receipt of the selection and/or new security value, may cause the security value to be reset, allowing use of the infusion system or pump **122**.

[0104] As an example, the safety system **204** may include a primary switch coupled to the supervisor chip **214** and a secondary switch coupled to the supervisor chip **214**. The primary switch may include the tamper switch **206**, or another safety system **204**, and the secondary switch may include the damage switch **208**, or another safety system **204**. When the supervisor chip **214** receives the repair data (or an indication that one or both of the primary and secondary switches are activated) from the primary and secondary switches, the supervisor chip **214** may record a timestamp associated with the activation of the primary and secondary switches and/or erase a security value from the memory of the supervisor chip, preventing use of the infusion system or pump. Accordingly, the implementation of the supervisor chip **214** with at least one or two other safety systems **204**, such as the tamper switch **206** and/or the damage switch **208**, may help to track unauthorized maintenance and/or repairs of the infusion system **120**, and may improve the safety of patients.

[0105] Referring again to FIG. 2, the pump **122** may include the tamper evident data tag **216**. The tamper evident data tag **216** may include a smart label, an RFID tag, a holographic tag, and/or the like. The tamper evident data tag **216** may be scanned when the housing **270** is appropriately positioned on the pump **122** and the pump **122** is properly assembled. When the pump **122** is accessed, such as during a repair event, the tamper evident data tag **216** may break, leaving it unable to be scanned. In this configuration, the repair data includes the no detection of the tamper evident data tag **216** when scanned (e.g., by a scanner of the pump or receipt of a scan by the safety engine **110**). The safety engine **110** may detect the repair event is unauthorized based on the no detection of the tamper evident data tag **216**, and cause at least one safety adjustment described herein.

[0106] In some implementations, the pump **122** includes the light sensor (e.g., switch) **218**. The light sensor **218** may be positioned within the housing **270** of the pump **122** and/or may be coupled to the pump **122** or infusion system **120**. The light sensor **218** may be used to identify when the housing **270** is removed from the pump **122** during a repair event. For example, the light sensor **218** may detect darkness, indicating the presence of the housing **270**. The light sensor **218** detecting light such that the amount of detected light exceeds a threshold amount indicates that the housing **270** has been removed. In some implementations, the light sensor **218** detecting the light causes the light sensor **218** and/or a switch coupled to the light sensor **218** to be thrown. When the light sensor **218** and/or switch is thrown, an indicator is written to the data storage **272**. In this instance, the repair data includes a scan of a memory coupled to the light sensor and/or switch to determine whether an event indicating the light sensor and/or switch was triggered is stored in the memory. In such configurations, the safety engine **110** may perform a startup procedure during which the safety engine **110** scans the data storage **272** for the indicator. If the safety engine **110** detects the indicator, the safety engine **110** may detect an unauthorized repair event, and as a result, cause at

least one safety adjustment, as described herein.

[0107] In some implementations, the pump **122** includes the camera **220**. The camera **220** may capture one or more images, videos, and/or the like. The camera **220** may be coupled to the display **236**. In some implementations, the safety engine **110** can determine, based on the captured images (e.g., the repair data), whether a repair event is occurring and/or whether the repair event is unauthorized. For example, the safety engine **110** may detect a repair event (e.g., an unauthorized repair event when a light level in a field of view of the camera corresponds to a light level threshold or an image shows total darkness. The light level in the field of view of the camera can indicate that the housing **270** and/or display **236** is face down on a surface or that the housing **270** is properly positioned on the pump **122** and the pump **122** is not being accessed. In some implementations, total darkness in the image can indicate that the housing **270** and/or display **236** is face down on a surface, while an image that is not total darkness can indicate that the housing **270** is properly positioned on the pump **122** and the pump **122** is not being accessed. In these configurations, the repair data may include a light level in a field of view of the camera, and the safety engine **110** detects an unauthorized repair event based on the light level corresponding to or not corresponding to a light level threshold. The safety engine **110** may as a result cause at least one safety adjustment, as described herein.

[0108] In some implementations, the pump **122** includes the drop sensor **222**. The drop sensor **222** may detect a pattern of movement representing a user accessing the pump **122** and/or making a repair on the pump **122**. The pattern may include an inversion or a quantity of inversion over a period of time. The safety engine **110** may receive the detected pattern, the inversion, or the quantity of inversion over the period of time (e.g., the repair data). Based on the received repair data, the safety engine may classify movement types. For example, the safety engine **110** may classify the movement type as a drop, an ambulatory transport, a repair, or another type of access of the pump **122**. If the safety engine **110** determines the movement represents a repair or another type of unauthorized access, the safety engine **110** may cause at least one safety adjustment, as described herein.

[0109] In some implementations, the patient safety device maintenance system **100** may help to prevent or limit diversion of the infusion system **120** (e.g., the pump **122**). For example, the infusion system **120** may be authorized and/or cleared for use in a certain location (e.g., jurisdiction). Diversion of the infusion system **120** may occur when the infusion system **120** is used in a location that is not authorized. If the infusion system **120** is used in a location that is not authorized, the safety engine **110** may determine that the use (e.g., during a repair event) of the infusion system **120** is unauthorized. As shown in FIG. 2, the safety system **204** may include the location sensor **224**. The location sensor **224** may determine a location of the pump **122** based on a GPS signal, a network address, a mobile identifier of a device communicatively coupled to the pump **122**, and/or the like. The safety engine **110** may receive such geofencing information as repair data, which the safety engine **110** may use to determine that the use of the pump is unauthorized. Based on the detection of the use in an unauthorized location, the safety engine **110** may cause at least one safety adjustment, as described herein.

[0110] FIG. 4A depicts a flowchart illustrating a process for safe repair of an infusion device, in accordance with some example implementations. Referring to FIG. 4A, the process **400** may be performed by the patient safety device maintenance system **100**, such as by the safety engine **110** or another component of the infusion system **120** (e.g., the pump **122**).

[0111] At **402**, the infusion system may activate a sensor (e.g., the tamper switch **206**, the damage switch **208**, the supervisor chip **214**, the tamper evident data tag **216**, the light sensor **218**, the camera **220**, the drop sensor **222**, the location sensor **224**, or the like) to detect repair events for the infusion system. The activation may include activating the safety system **204** (e.g., one or more of the sensors, switches, accelerometers, scanners, etc. as described) to determine when the infusion system is accessed for a repair. In some implementations, the activation may be based on detecting

an event such as orientation of the infusion system, a change in orientation of the infusion system, opening of a housing of the infusion system, removal of a component from the housing of the infusion system, activation of a button or other control element to indicate a repair mode, or the like.

[0112] At **404**, the sensor may collect repair data for a detected repair event. The repair data may include user information for a user performing the repair, part information for a part used or replaced during the repair, or other information as described. Based at least in part on the repair data, at **406**, the safety engine or the infusion system may determine whether the repair event is completed. The determination may include determining a sequence of sensor readings corresponds to a repair sequence. The determination may include assessing a measurement, such as the repair data or other measurements, of the sensor. The determination may include receiving a user input via a graphical user interface indicating the repair is complete. The determination may include detecting removal of a user token (e.g., badge of the user moving a certain distance away from the infusion system, disconnection of a user token from the infusion system, disconnection of a data path between the infusion system and a maintenance device, or the like).

[0113] Collecting the repair data may include transmitting the repair data to a safety server or other network device. In some implementations, the collection of repair data may be performed without indication to the user performing the repairs. In some implementations, it may be desirable to randomly collect repair data. In this way, a repair user may not know whether a particular repair is being monitored or not.

[0114] If the determination at **406** is negative, the process **400** may return to block **404** to collect additional repair data. Based on the repair being performed, the collection may include adjusting the sensor(s) used to collect the repair data. For example, once the housing is opened, it may not be necessary to monitor the state of a switch to detect access into the housing but rather shift to detecting, for example, battery state. In this way, the infusion system may conserve resources needed to collect and process the repair data by focusing resources on the parts or sensors related to an anticipated step in the repair process. The steps may be identified using a list or look up table where each entry identifies the particular sensor to activate and, in some implementations, a configuration of the sensor (e.g., light level threshold, load to detect, voltage threshold, target orientation, etc.).

[0115] Returning to **406**, if the repair event is complete, the process **400** may proceed to **408**. At **408**, the safety engine or safety server (e.g., the server **126**) may determine whether the repair event was authorized. The determination may be performed based on the repair data as described. If affirmative, the process **400** may moves to **412** to record authorization of the repair event. Recording may include storing a flag in a secure memory of the infusions system indicating the device is in an authorized state of repair. Recording may include submitting an entry to a distributed ledger or device registry to indicate the repair event. After recording, the process **400** may end.

[0116] Returning to **408**, if the repair event is unauthorized, the process **400** at **410** may record an unauthorized repair event for the infusion system. Recording may include storing a flag in a secure memory of the infusions system indicating the device is in an unauthorized state of repair. Recording may include submitting an entry to a distributed ledger or device registry to indicate the repair event. After recording, the process **400** may end.

[0117] FIG. **4B** depicts a flowchart illustrating a process **450** for improving patient safety, consistent with implementations of the current subject matter. Referring to FIG. **4B**, the process **450** may be performed by the patient safety device maintenance system **100**, such as by the safety engine **110** or another component of the infusion system **120** (e.g., the pump **122**).

[0118] At **452**, the infusion system may be initiated for operation. Initiation may include adjusting the power state of the infusion system (e.g., power on, activate from stand-by, etc.). Initiation may include receiving an automated programming request to administer an infusion by the infusion

system. In some implementations, initiation may include activation of a control element on a user interface of the infusion system.

[0119] At **454**, the infusion system may receive repair data from the safety engine **110**, or directly from the server **126**, coupled to an infusion system (e.g., the infusion system **120**). The repair data may be associated with a repair event, during which a user accesses the infusion system. In some implementations, the repair data may be processed by the safety engine to indicate whether the associated repair event was authorized. In some implementations, the infusion system may receive the repair data and determine authorization by processing the repair data. For example, where the repair data for the infusion system is added to a distributed ledger, the infusion system may access ledger entries associated with the infusion system. The infusion system may search the ledger based on date. For example, the infusion system may query the ledger of entries occurring after the date of the last initiation. In this way, only records that have not been reviewed are received. This can reduce the resources and time needed to determine the authorization state of the infusion system. In some implementations, the ledger may be searched using an identifier of the infusion system. This may further reduce the number of records reviewed and corresponding resources used to determine the authorization state of the infusion system.

[0120] Consistent with implementations of the current subject matter, the repair event may include a repair of the infusion system, a maintenance performed on the infusion system, use of the infusion system, and/or the like. The repair event may be authorized or unauthorized. The repair event is authorized when the repair event is performed by an authorized user (e.g., a certified, trained, and/or qualified user). Additionally or alternatively, the repair event is authorized when an authorized component (e.g., a manufacturer approved component) is used to repair and/or replace a component of the infusion system. The repair event may be unauthorized when the repair event is performed by an unauthorized user (e.g., a user that is not certified, trained, and/or qualified). The repair event may also be unauthorized when an unauthorized component, such as a component that has not been approved by the manufacturer, is used to repair and/or replace a component of the infusion system.

[0121] The safety engine **110** may receive the repair data from the safety system. The safety system may include one or more switches, sensors, processors, and/or the like, that are used to generate repair data for use by the safety engine **110** to detect authorized and/or unauthorized access of the infusion system, such as the pump. For example, the safety system may include one or more of a tamper switch (e.g., the tamper switch **206**), a damage switch (e.g., the damage switch **208**), a supervisor chip (e.g., the supervisor chip **214**), a tamper evident data tag (e.g., the tamper evident data tag **216**), a light sensor (e.g., the light sensor **218**), a camera (e.g., the camera **220**), a drop sensor (e.g., the drop sensor **222**), a location sensor (e.g., the location sensor **224**), and/or the like. The repair data generated by the safety system may include an activation of the safety system (or an indication of the activation), captured and/or measured data including timestamps, images, acceleration data, moisture data, light data, and/or the like.

[0122] At **456**, the safety engine **110** determines whether the repair event is authorized based at least in part on the repair data. For example, the safety engine **110** may compare the repair data to a threshold and detect the repair event is unauthorized based on the repair data corresponding to the threshold. The safety engine **110** may additionally or alternatively determine that the activation of one or more sensors indicates that the repair event is unauthorized. The safety engine **110** may additionally or alternatively detect the repair event is unauthorized based on the presence or absence of a security value. The safety engine **110** may additionally or alternatively detect the repair event is unauthorized based on the location of the infusion system, and/or the like. If authorized, the process **450** ends but may be repeated when the infusion system is initialized at a different time.

[0123] If unauthorized, the process **450** may move to **458**. At **458**, the safety engine **110** causes at least one safety adjustment to operation of the infusion system based on the detection that the repair



event is unauthorized. Additionally or alternatively, the safety engine **110** causes at least one safety adjustment based on detection of a repair event. The at least one safety adjustment may include transmission of an alert, storage of information associated with the repair event, transmission of an alarm, disablement of the infusion system, changing an operation of the infusion system, and/or the like. Thus, the safety engine **110** may track access of the infusion system for later auditing and/or investigation, and in some instances, may prevent use of the infusion system until the infusion system **120** is evaluated by an authorized user. Accordingly, the safety engine **110** may improve patient safety and allow for tracking of repair events, including unauthorized repair events.

[0124] FIG. 5 depicts a block diagram illustrating a computing system **500** consistent with implementations of the current subject matter. Referring to FIGS. 1 and 5, the computing system **500** can be used to implement the patient safety device maintenance system **100**, the pump **122**, the safety engine **110** and/or any components therein.

[0125] As shown in FIG. 5, the computing system **500** can include a processor **510**, a memory **520**, a storage device **530**, and input/output devices **540**. The processor **510**, the memory **520**, the storage device **530**, and the input/output devices **540** can be interconnected via a system bus **550**. The processor **510** is capable of processing instructions for execution within the computing system **500**. Such executed instructions can implement one or more components of, for example, the safety engine **110**. In some example implementations, the processor **510** can be a single-threaded processor. Alternatively, the processor **510** can be a multi-threaded processor. The processor **510** is capable of processing instructions stored in the memory **520** and/or on the storage device **530** to present graphical information for a user interface provided via the input/output device **540**.

[0126] The memory **520** is a computer readable medium such as volatile or non-volatile that stores information within the computing system **500**. The memory **520** can store data structures representing configuration object databases, for example. The storage device **530** is capable of providing persistent storage for the computing system **500**. The storage device **530** can be a floppy disk device, a hard disk device, an optical disk device, or a tape device, or other suitable persistent storage means. The input/output device **540** provides input/output operations for the computing system **500**. In some example implementations, the input/output device **540** includes a keyboard and/or pointing device. In various implementations, the input/output device **540** includes a display unit for displaying graphical user interfaces.

[0127] According to some example implementations, the input/output device **540** can provide input/output operations for a network device. For example, the input/output device **540** can include Ethernet ports or other networking ports to communicate with one or more wired and/or wireless networks (e.g., a local area network (LAN), a wide area network (WAN), the Internet).

[0128] In some example implementations, the computing system **500** can be used to execute various interactive computer software applications that can be used for organization, analysis and/or storage of data in various formats. Alternatively, the computing system **500** can be used to execute software applications. These applications can be used to perform various functionalities, e.g., planning functionalities (e.g., generating, managing, editing of spreadsheet documents, word processing documents, and/or any other objects, etc.), computing functionalities, communications functionalities, etc. The applications can include various add-in functionalities or can be standalone computing products and/or functionalities. Upon activation within the applications, the functionalities can be used to generate the user interface provided via the input/output device **540**. The user interface can be generated and presented to a user by the computing system **500** (e.g., on a computer screen monitor, etc.).

[0129] In some example implementations, the pump **122** may be part of a patient care system **20**. FIGS. 6A-6C illustrate example implementations of the patient care system **20**, though other types of patient care systems may be implemented. Referring to FIG. 6A, the patient care system **20** may include the pump **122** as well as additional pumps **24**, **26**, and **28**. Although a large volume pump (LVP) is illustrated, other types of pumps may be implemented, such as a small volume pump

(SVP), a TCI pump, a syringe pump, an anesthesia delivery pump, and/or a patient-controlled analgesic (PCA) pump configured to deliver a medication (e.g., an anesthesia, such as propofol and/or remifentanyl, and the like) to a patient. The pump **122** may be any infusion device configured to deliver a substance (e.g., fluid, nutrients, medication, and/or the like) to a patient's circulatory system or epidural space via, for example, intravenous infusion, subcutaneous infusion, arterial infusion, epidural infusion, and/or the like, or the pump **122** may be an infusion device configured to deliver a substance (e.g., fluid, nutrients, medication, and/or the like) to a patient's digestive system via a nasogastric tube (NG), a percutaneous endoscopic gastrostomy tube (PEG), nasojugal tube (NJ), and/or the like.

[0130] As shown in FIG. **6A**, each of the pump **122**, **24**, **26**, and **28** may be fluidly connected with an upstream fluid line **30**, **32**, **34**, and **36**, respectively. Moreover, each of the four pumps **122**, **24**, **26**, and **28** may also fluidly connected with a downstream fluid line **31**, **33**, **35**, and **37**, respectively. The fluid lines can be any type of fluid conduit, such as tubing, through which fluid can flow. At least a portion of one or more of the fluid lines may be constructed with a multi-layered configuration as described herein.

[0131] Fluid supplies **38**, **40**, **42**, and **44**, which may take various forms but in this case are shown as bottles, are inverted and suspended above the pumps. Fluid supplies may also take the form of bags, syringes, or other types of containers. Both the patient care system **20** and the fluid supplies **38**, **40**, **42**, and **44** may be mounted to a roller stand or intravenous (IV) pole **46**.

[0132] A separate pump **122**, **24**, **26**, and **28** may be used to infuse each of the fluids of the fluid supplies into the patient. The pumps **122**, **24**, **26**, and **28** may be flow control devices that will act on the respective fluid line to move the fluid from the fluid supply through the fluid line to the patient **48**. Because individual pumps are used, each can be individually set to the pumping or operating parameters required for infusing the particular medical fluid from the respective fluid supply into the patient at the particular rate prescribed for that fluid by the physician. Such medical fluids may comprise drugs or nutrients or other fluids.

[0133] Typically, medical fluid administration sets have more parts than are shown in FIG. **6A**. Many have check valves, drip chambers, valved ports, connectors, and other devices well known to those skilled in the art. These other devices have not been included in the drawings so as to preserve clarity of illustration. In addition, it should be noted that the drawing of FIG. **6A** is not to scale and that distances have been compressed for the purpose of clarity. In an actual setting, the distance between the bottles **38**, **40**, **42**, and **44** and the pump modules **22**, **24**, **26**, and **28** could be much greater.

[0134] Referring now to FIG. **6B**, an enlarged view of the front of the patient care system **20** is shown. The pump **122** may include a front door **50** and a handle **52** that operates to lock the door in a closed position for operation and to unlock and open the door for access to the internal pumping and sensing mechanisms and to load administration sets for the pump. In some implementations, the safety engine **110** determines when the door, the handle, and/or other components of the pump **122** are accessed, replaced, and/or repaired. When the door is open, the tube can be connected with the pump, as will be shown in FIG. **6C**. When the door is closed, the tube is brought into operating engagement with the pumping mechanism, the upstream and downstream pressure sensors, and the other equipment of the pump. A display **154**, such as an LED display, is located in plain view on the door in this embodiment and may be used to visually communicate various information relevant to the pump, such as alert indications (e.g., alarm messages). For example, the display **154** may display one or more alerts and/or indications indicating access, such as authorized and/or unauthorized access of the infusion system. The display **154** may otherwise be a part of or be coupled to the pump **122**. Control keys **56** exist for programming and controlling operations of the pump as desired. The pump **122** also includes audio alarm equipment in the form of a speaker (not shown).

[0135] In the embodiment shown, a programming module **60** is attached to the left side of the

pump **122**. In some implementations, the programming module **60** forms a part of the pump **122**. Other devices or modules, including another pump, may be attached to the right side of the pump **122**, as shown in FIG. **6A**. In such a system, each attached pump represents a pump channel of the overall patient care system **20**. In one embodiment, the programming module is used to provide an interface between the pump **122** and external devices as well as to provide most of the operator interface for the pump **122**.

[0136] The programming module **60** includes a display **62** for visually communicating various information, such as the operating parameters of the pump **122** and alert indications and alarm messages. The programming module **60** may also include a speaker to provide audible alarms. The programming module or any other module also has various input devices in this embodiment, including control keys **64** and a bar code or other scanner or reader for scanning information from an electronic data tag relating to the infusion, the patient, the care giver, or other. The programming module also has a communications system (not shown) with which it may communicate with external equipment such as a medical facility server or other computer and with a portable processor, such as a handheld portable digital assistant (“PDA”), or a laptop-type of computer, or other information device that a care giver may have to transfer information as well as to download drug libraries to a programming module or pump. In some implementations, the programming module **60** may communicate with the safety engine **110**, include the safety engine **110**, or implement features of the safety engine **110** described herein.

[0137] The communications system may take the form of a radio frequency (“RF”) (radio frequency) system, an optical system such as infrared, a Blue Tooth system, or other wired or wireless system. The bar code scanner and communications system may alternatively be included integrally with the pump **122**, such as in cases where a programming module is not used, or in addition to one with the programming module. Further, information input devices need not be hard-wired to medical instruments, information may be transferred through a wireless connection as well.

[0138] FIG. **6B** includes a second pump **26** connected to the programming module **60**. As shown in FIG. **6A**, more pump modules may be connected. Additionally, other types of modules may be connected to the pump modules or to the programming module.

[0139] Turning now to FIG. **6C**, the pump **122** is shown in perspective view with the front door **50** open, showing the upstream fluid line **30** and downstream fluid line **31** in operative engagement with the pump **122**. The pump **122** directly acts on a tube **66** (also referred to as a pump segment) that connects the upstream fluid line **30** to the downstream fluid line **31** to form a continuous fluid conduit, extending from the respective fluid supply **38** (FIG. **6A**) to the patient **48**, through which fluid is acted upon by the pump to move fluid downstream to the patient. Specifically, a pumping mechanism **70** acts as the flow control device of the pump to move fluid through the conduit. The upstream and downstream fluid lines and/or tube **66** may be coupled to a pump cassette or cartridge that is configured to be coupled to the pump **122**, such as the type described in co-pending U.S. patent application Ser. No. 13/827,775, which is incorporated by reference herein.

[0140] The type of pumping mechanism may vary and may be for example, a multiple finger pumping mechanism. For example, the pumping mechanism may be of the “four finger” type and includes an upstream occluding finger **72**, a primary pumping finger **74**, a downstream occluding finger **76**, and a secondary pumping finger **78**. The “four finger” pumping mechanism and mechanisms used in other linear peristaltic pumps operate by sequentially pressing on a segment of the fluid conduit by means of the cam-following pumping fingers and valve fingers **72**, **74**, **76**, and **78**. The pressure is applied in sequential locations of the conduit, beginning at the upstream end of the pumping mechanism and working toward the downstream end. At least one finger is always pressing hard enough to occlude the conduit. As a practical matter, one finger does not retract from occluding the tubing until the next one in sequence has already occluded the tubing; thus at no time is there a direct fluid path from the fluid supply to the patient. The operation of peristaltic pumps

including four finger pumps is well known to those skilled in the art and no further operational details are provided here.

[0141] In this particular embodiment, FIG. 6C further shows a downstream pressure sensor **82** included in the pump **122** at a downstream location with respect to the pumping mechanism. The downstream pressure sensor **82** is mounted to the flow control device **70** and is located adjacent and downstream in relation to the flow control device. The downstream pressure sensor is located downstream from the flow control device, that is, at a location between the patient **48** (FIG. 6A) and the flow control device, so that the connection of the correct fluid supply with the correct pump may be verified before any fluid is pumped to the patient.

[0142] With reference still to FIG. 6C, an upstream pressure sensor **80** may also be included in the pump **122**. The upstream pressure sensor is assigned to the flow control device or pumping mechanism **70** and, in this embodiment, is further provided as an integral part of the pump **122**. It is mounted to the flow control device **70** and is located adjacent and upstream in relation to the flow control device. The upstream pressure sensor is located upstream from the flow control device, that is, at a location between the fluid supply **38** (FIG. 6A) and the flow control device, so that the connection of the correct fluid supply with the correct pump may be verified before any fluid is pumped to the patient. In an implementation where the source is a syringe, the flow control device **70** may be configured to press a plunger of the syringe to provide the infusion according to the programmed parameters.

[0143] FIG. 7 depicts a process diagram for secure repair of a device, in accordance with some example implementations. The process **700** may include a user **702** to repair a device **708**, which may be and/or include the infusion system **120**. The user **702** may initiate repair (e.g., a repair event). Initiating repair may include activating a user interface element of the device **708**. Initiating repair may include opening the housing of the device **708**. Initiating repair may include virtually accessing firmware or other software stored in memory of the device **708**.

[0144] After initiating repair, the user **702** may provide a user token **704** to the device **708**. As shown in FIG. 7, the device **708** may include a scanner **710**. The scanner **710** may be activated to receive the user token **704**. The user token **704** may be a wireless contact card including information that can be read by the scanner **710**. The user token **704** may include biometric information such as a fingerprint, facial recognition, iris scan, or the like. In some implementations, the user token **704** may be stored in a storage device such as a removable storage drive that can be connected to the device **708**. The user authorization via the user token **704** allows the system to associate activities and associated repair data at the device **708** with the user **702** associated with the user token **704**.

[0145] In some implementations, the user **702** may repair, replace, update, or clean a part of the device **708**. The user **702** may scan the new part using the scanner **710**. The user **702** may additionally or alternatively scan the old part using the scanner **710**. In this way, a record of the part(s) included in the device **708** may be recorded as described herein (e.g., blockchain, database, distributed ledger, etc.).

[0146] The device **708** may confirm that the repair is performed or completed. The confirmation may include determining the housing is secured or receiving a completion indication via a user interface. In some implementations, the scanner **710** may activate after the housing is secure to take an inventory of parts within the housing of the device **708**. This inventory information may be included in the repair data associated with the repair event.

[0147] The device **708** may transmit the repair data to a safety engine **712**. The transmission may include an identifier for the device **708** such as a media access control identifier, network address, serial number, or other information to uniquely identify the device **708**. The safety engine **712** may store the repair data in a data store, such as at a server. In some implementations, the safety engine **712** may process the repair data to assess the repair data to determine whether the repair is authorized or unauthorized. The assessment may be performed in real-time, during the repair event.

As used herein, “real-time” may refer to availability for processing at or near in time to the time the associated repair data is generated or detected. In some implementations, the assessment may be performed after the repair event is complete. For example, when the device **708** is powered on after the repair event, the initialization of the device **708** may include communication with the safety engine **712** to determine whether the device **708** was improperly repaired. As discussed, in some implementations, the safety engine **712** may be included in the device **708**.

[0148] FIG. **8** depicts a state diagram of user interfaces for improving patient safety when using a repaired infusion device, in accordance with some example implementations. At state **808**, the device may indicate a readiness to infuse. From state **808**, the device may be explicitly activated to enter a repair mode, state **802**. The activation may include selecting a control element on a user interface or pressing a specific button or sequence of buttons to enter a maintenance/repair mode. Once in state **802**, the device may return to state **808** if the user cancels the repair mode. Cancelling the repair mode without making any changes may be desirable when the repair mode is accidentally initiated or detected (e.g., based on orientation, location, etc.). However, if the repair mode is confirmed (e.g., selecting the YES button), the device may transition to state **804**. At state **804**, an indication of the state of the device as being in “repair mode” may be provided. The indication may be provided, for example, as a static element on the user interface of the device, such as shown in FIG. **8**. Other manifestations may include activating a light, sound, or other human perceivable indication of the repair state.

[0149] From state **804**, a user may continue repairing the device. The continue control element may be presented in implementations where the user performs a series of steps to make the repair such as opening the housing, scanning a new part, etc. If the repair is continued, the device may remain in state **804**. However, if the repair is marked complete, the device may move to either state **806** or state **808** depending on whether the repair event was authorized.

[0150] To move to state **808**, the security engine may process the repair event and associated repair data to determine whether the repair was authorized. Authorization may include confirming, via the safety engine as described, the permissions of the user associated with the repair, the parts used during the repair, the order of steps performed for the repair, the operational status of the device after the repair, or the like as derived from the repair data and/or device data (e.g., self-test, switch state, value stored at a secured memory location, etc.). If the repair event was authorized, the device may transition to state **808**. From state **808**, the device may be operated with confidence that the repaired state was performed safely. From state **808**, the device may transition to state **802** to perform another repair as described.

[0151] However, from state **804**, if the repair event was unauthorized, the device may transition to state **806**. In the example shown in FIG. **8**, the user interface may present a perceivable indicator that the device was repaired in an unauthorized fashion. The indication may be provided, for example, as a static element on the user interface of the device, such as shown in FIG. **8**. Other manifestations may include activating a light, sound, or other human perceivable indication of the repair state.

[0152] As shown in FIG. **8**, the device in state **806** may provide some infusion functionality. For example, the device may disable network connectivity to prevent malicious or unsecured software from the clinical network. However, the device may still be accessed and used to provide treatment to a patient. In some implementations, the device at state **806** may provide an option to verify an unauthorized repair. As shown, the user interface of the device may include a control element (e.g., button) to verify the repair. Once activated, the device may require one or more input values to allow an authorized user to confirm the repair and return the device to the authorized state **808**. The confirmation may include receiving a user token or other credential to “sign-off” on the device.

[0153] In some implementations, the device may include two or more repair events. In such instances, one repair event may be authorized while a second repair event may be unauthorized. While in the unauthorized state **806**, the device may be operated to return to state **802** to perform an

additional repair. If the additional repair is not performed (e.g., cancelled) the device will return to the unauthorized state **806**.

[0154] One or more aspects or features of the subject matter described herein can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs, field programmable gate arrays (FPGAs) computer hardware, firmware, software, and/or combinations thereof. These various aspects or features can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which can be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device. The programmable system or computing system may include clients and servers. A client and server are remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

[0155] These computer programs, which can also be referred to as programs, software, software applications, applications, components, or code, include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the term “machine-readable medium” refers to any computer program product, apparatus and/or device, such as for example magnetic discs, optical disks, memory, and Programmable Logic Devices (PLDs), used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor. The machine-readable medium can store such machine instructions non-transitorily, such as for example as would a non-transient solid-state memory or a magnetic hard drive or any equivalent storage medium. The machine-readable medium can alternatively or additionally store such machine instructions in a transient manner, such as for example, as would a processor cache or other random access memory associated with one or more physical processor cores.

[0156] To provide for interaction with a user, one or more aspects or features of the subject matter described herein can be implemented on a computer having a display device, such as for example a cathode ray tube (CRT) or a liquid crystal display (LCD) or a light emitting diode (LED) monitor for displaying information to the user and a keyboard and a pointing device, such as for example a mouse or a trackball, by which the user may provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well. For example, feedback provided to the user can be any form of sensory feedback, such as for example visual feedback, auditory feedback, or tactile feedback; and input from the user may be received in any form, including acoustic, speech, or tactile input. Other possible input devices include touch screens or other touch-sensitive devices such as single or multi-point resistive or capacitive track pads, voice recognition hardware and software, optical scanners, optical pointers, digital image capture devices and associated interpretation software, and the like.

[0157] In the descriptions above and in the claims, phrases such as “at least one of” or “one or more of” may occur followed by a conjunctive list of elements or features. The term “and/or” may also occur in a list of two or more elements or features. Unless otherwise implicitly or explicitly contradicted by the context in which it is used, such a phrase is intended to mean any of the listed elements or features individually or any of the recited elements or features in combination with any of the other recited elements or features. For example, the phrases “at least one of A and B;” “one or more of A and B;” and “A and/or B” are each intended to mean “A alone, B alone, or A and B together.” A similar interpretation is also intended for lists including three or more items. For example, the phrases “at least one of A, B, and C;” “one or more of A, B, and C;” and “A, B, and/or C” are each intended to mean “A alone, B alone, C alone, A and B together, A and C together, B

and C together, or A and B and C together.” Use of the term “based on,” above and in the claims is intended to mean, “based at least in part on,” such that an unrecited feature or element is also permissible.

[0158] As used herein a “user interface” (also referred to as an interactive user interface, a graphical user interface or a UI) may refer to a network based interface including data fields and/or other control elements for receiving input signals or providing electronic information and/or for providing information to the user in response to any received input signals. Control elements may include dials, buttons, icons, selectable areas, or other perceivable indicia presented via the UI that, when interacted with (e.g., clicked, touched, selected, etc.), initiates an exchange of data for the device presenting the UI. A UI may be implemented in whole or in part using technologies such as hyper-text mark-up language (HTML), FLASH™, JAVA™, .NET™, web services, or rich site summary (RSS). In some implementations, a UI may be included in a stand-alone client (for example, thick client, fat client) configured to communicate (e.g., send or receive data) in accordance with one or more of the aspects described. The communication may be to or from a medical device or server in communication therewith.

[0159] The subject matter described herein can be embodied in systems, apparatus, methods, and/or articles depending on the desired configuration. The implementations set forth in the foregoing description do not represent all implementations consistent with the subject matter described herein. Instead, they are merely some examples consistent with aspects related to the described subject matter. Although a few variations have been described in detail above, other modifications or additions are possible. In particular, further features and/or variations can be provided in addition to those set forth herein. For example, the implementations described above can be directed to various combinations and subcombinations of the disclosed features and/or combinations and subcombinations of several further features disclosed above. In addition, the logic flows depicted in the accompanying figures and/or described herein do not necessarily require the particular order shown, or sequential order, to achieve desirable results. Other implementations may be within the scope of the following claims.

## Claims

1. A system, comprising: at least one data processor; and at least one memory storing instructions which, when executed by the at least one data processor, result in operations comprising: receiving, from a safety system coupled to an infusion system, repair data associated with a repair event, during which an element within a housing of the infusion system is accessed; detecting, based on the repair data, the repair event is unauthorized; and causing, based on the detecting, at least one safety adjustment to operation of the infusion system.
2. The system of claim 1, wherein the repair event comprises one or more of a repair of the element of the infusion system, a replacement of the element of the infusion system, opening the housing, and a maintenance performed on the infusion system.
3. The system of claim 1, wherein the infusion system comprises an infusion pump.
4. The system of claim 1, wherein the infusion system comprises: an infusion pump; and a control device coupled to the infusion pump, the control device comprising at least one data processor and at least one memory storing instructions which, when executed by the at least one data processor, control operation of the infusion pump.
5. The system of claim 1, wherein the safety system comprises a tamper switch, the tamper switch comprising: at least two electrically conductive surfaces of the infusion system; wherein contact between the at least two electrically conductive surfaces forms a circuit; and a fastener configured to maintain contact between the at least two electrically conductive surfaces; wherein removal of the fastener is configured to allow access to the infusion system during the repair event; and wherein the removal of the fastener is further configured to activate the tamper switch.

6. The system of claim 5, wherein the safety system further comprises the housing of the infusion system, the housing defining at least one of the at least two electrically conductive surfaces.
7. The system of claim 6, wherein the repair data comprises information indicating activation of the tamper switch; and wherein the detecting comprises determining the repair event is unauthorized when the tamper switch is activated.
8. The system of claim 1, wherein the safety system comprises a damage switch configured to indicate the infusion system should be accessed during the repair event, wherein the damage switch comprises one or more of a moisture sensor configured to be activated upon detecting an ingress of fluid into an interior of the infusion system and an accelerometer configured to be activated upon detecting an acceleration indicating an impact on the infusion system.
9. (canceled)
10. (canceled)
11. The system claim 1, wherein the safety system comprises: a supervisor chip coupled to the infusion system; a primary switch coupled to the supervisor chip and to the infusion system; and a secondary switch coupled to the supervisor chip and to the infusion system.
12. (canceled)
13. The system of claim 11, wherein the repair data comprises information indicating activation of one or more of the primary switch and the secondary switch.
14. The system of claim 13, wherein the at least one safety adjustment comprises recording, by the supervisor chip, a timestamp associated with the activation of the one or more of the primary switch and the secondary switch.
15. The system of claim 13, wherein the at least one safety adjustment comprises erasing a security value from memory of the supervisor chip, thereby limiting use of the infusion system.
16. The system of claim 15, wherein the operations further comprise: resetting the security value upon receipt of a selection from an authorized user, thereby restoring the use of the infusion system.
17. (canceled)
18. The system claim 1, wherein the safety system comprises a light sensor and/or switch within a housing of the infusion system; wherein the repair data comprises a scan of a memory coupled to the light sensor and/or switch to determine whether an event indicating the light sensor and/or switch was triggered is stored in the memory, the light sensor and/or switch being triggered when the housing is removed from the infusion system; and wherein the detecting comprises determining the repair event is unauthorized based on a determination that the event is stored in the memory.
19. The system claim 1, wherein the safety system comprises a camera coupled to the housing of the infusion system; wherein the repair data comprises a light level in a field of view of the camera; and wherein the detecting comprises determining the repair event is unauthorized based on the light level in the field of view of the camera corresponding to a light level threshold.
20. (canceled)
21. The system claim 1, further comprising an authorized user registry; and wherein the repair data includes user data associated with the infusion system during the repair event, and wherein determining the repair event is unauthorized includes determining that the user data is unassociated with a user included in the authorized user registry.
22. The system claim 1, further comprising an authorized user registry; and wherein the repair data includes user data associated with the infusion system during the repair event, and wherein determining the repair event is unauthorized includes determining that the user data is associated with a user authorized to perform a different repair event than the repair event in the authorized user registry.
23. (canceled)
24. (canceled)
25. (canceled)



**26.** (canceled)

**27.** (canceled)

**28.** (canceled)

**29.** (canceled)

**30.** (canceled)

**31.** The system claim 1, wherein the at least one safety adjustment comprises creation, in a distributed ledger, of an entry comprising an identifier associated with the infusion system and a timestamp of the repair event.

**32.** The system of claim 31, wherein the distributed ledger is a blockchain ledger.

**33.** (canceled)

**34.** (canceled)

**35.** (canceled)

**36.** A method, comprising: receiving, from a safety system coupled to an infusion system, repair data associated with a repair event, during which an element within a housing of the infusion system is accessed; detecting, based on the repair data, the repair event is unauthorized; and causing, based on the detecting, at least one safety adjustment to operation of the infusion system.

**37.** The method of claim 36, wherein the repair event comprises one or more of a repair of the element of the infusion system, a replacement of the element of the infusion system, opening the housing, and a maintenance performed on the infusion system.

**38-81.** (canceled)

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